Blessed G/Michael Catholic Secondary School
1st Semester Chemistry Test One for Grade 12th

NOV, 2018 E.C Time Allowed:1:20 hr

Name: Hanna Ayalew **Sec:**_____ **R.No.**_____

Choose the best answer from the given alternatives

_____1. A solution in which [H+] = 10-8 M has a pH of __ and is __.

A. 8, acidic B. 6, basic C. 8, basic D. -6, basic

_____2. Which of the following statements best explains why NH_3 is *not* an Arrhenius base, although it forms OH^- ions in water?

A) It does not contain OH^- in its formula.
B) It is a weak electrolyte.
C) It does not directly release OH^- ions but reacts with water to form them.
D) It acts only in non-aqueous media.

_____3. In the reaction $H_2PO_4^- \rightarrow HPO_4^{2-} + H^+$, the conjugate acid-base pair is:

A) $H_2PO_4^-$ / HPO_4^{2-} C) HPO_4^{2-} / H^+
B) $H_2PO_4^-$ / H^+ D) None of these

_____4. Which statement correctly compares acid and base strength in Brønsted theory?

A) Strong acids have weak conjugate bases.
B) Strong bases have strong conjugate acids.
C) Strong acids form strong conjugate bases.
D) Acid and conjugate base have equal strength.

_____5. Which of the following is not amphiprotic?

A) H_2O B) HSO4 - C) HCO3 - D) HNO3

_____6. The reaction $Ag^+ + 2NH_3 \rightarrow [Ag(NH_3)_2]^+$ is best explained by

A) Arrhenius concept C) Lewis concept
B) Brønsted–Lowry concept D) none of these

_____7. Identify the Lewis acid in the reaction:

$Fe^{3+} + 6H_2O \rightarrow [Fe(H_2O)_6]^{3+}$

A) Fe^{3+} B) H2O C) $[Fe(H_2O)_6]^{3+}$ D) OH^-

_____8. If the temperature increases, Kw increases. What happens to the pH of pure water?

A) pH increases C) pH remains 7
B) pH decreases D) Depends on the base used

9. Which of the following statements about the ionization constant of water is **incorrect?**

A) Kw increases with temperature because ionization is endothermic.
B) $K_w = [H^+][OH^-] = (10^{-7})^2$ at 25°C.
C) The neutral pH of water always equals 7 regardless of temperature.
D) In pure water, $[H^+] = [OH^-]$.

10. The stronger the acid:

A) The smaller the Ka and larger the pKa
B) The larger the Ka and smaller the pKa
C) The smaller both Ka and pKa
D) The larger both Ka and pKa

11. If pKa = 3 and pKb = 11 for a conjugate pair, Kw equals:
A) 10^{-8} B) 10^{-14} C) 10^{-4} D) 10^{-7}

12. If the percent ionization of a 0.10 M solution is 1.0%, Ka = ?

A) 1.0×10^{-5} B) 1.0×10^{-4} C) 1.0×10^{-3} D) 1.0×10^{-6}

13. Two acids, HA ($K_a=10^{-5}$) and HB ($K_a=10^{-7}$), both 0.1 M. Which is more ionized?

A) HA B) HB C) Both equal D) Cannot be compared

14. For a weak base with pKb = 4, what is the pKa of its conjugate acid?
A) 4 B) 10 C) 14 D) 18

15. NH_4Cl solution is acidic because:
A) Cl^- is hydrolyzed
B) NH_4^+ acts as a weak acid
C) NH_4Cl reacts with water to produce OH^-
D) Both ions undergo equal hydrolysis

16. The ionization of CH_3COOH is suppressed by adding:
A) NaOH B) $NaCH_3COO$ C) HCl D) NaCl

17. An acidic buffer is made by mixing:

A) Weak acid + its salt with a strong acid
B) Weak acid + its salt with a strong base
C) Strong acid + weak base
D) Weak base + its salt with a strong acid

18. For a buffer with [acid] = 0.005 M, [salt] = 0.5 M, Ka = 1.0×10^{-4}, find pH.
A) 4.00 B) 4.3 C) 6 D) 5

19. Which buffer will have pH closest to 9.0?

A) NH_4OH/NH_4Cl C) $H_2CO_3/NaHCO_3$
B) CH_3COOH/CH_3COONa D) NaH_2PO_4/Na_2HPO_4

20. An acid–base indicator is best described as:
A) A strong acid that changes color on dilution
B) A weak acid or base whose conjugate pair has different colors
C) A dye that absorbs UV light only
D) A salt that undergoes hydrolysis

21. The equivalence point in an acid–base titration is the point at which:
A) The pH = 7
B) The number of moles of acid = number of moles of base
C) The indicator changes color
D) The conductivity is maximum

22. For strong acid vs. weak base titration, pH at equivalence is:
A) <7 B) =7 C) >7 D) Cannot be predicted

23. The pH at the half-neutralization point equals:
A) 7 B) pKa of acid C) ½ (pKa + pKw) D) pKb of base

24. If 0.1 M NH_3 is titrated with 0.2 M HCl, the equivalence point occurs at what volume of acid (base volume = 50 mL)?
A) 10 mL B) 25mL C) 50mL D) 100 mL

25. For CH_3COOH/CH_3COONa buffer, adding strong base:
A) Decreases pH slightly C) No change
B) Increases pH slightly D) Drastically increases pH

26. The solution of known concentration in titration is called:

A) Analyte C) Standard solution or titrant
B) Indicator D) precipitate

27. Which of the following is the best explanation for why **calcium fluoride** (CaF_2) has reduced solubility in a solution containing **NaF**?

A. The fluoride ion (F^-) from NaF shifts the equilibrium of CaF_2 dissociation to the left
B. The sodium ion (Na^+) from NaF enhances the dissociation of CaF_2
C. The fluoride ion (F^-) from NaF shifts the equilibrium of CaF_2 dissociation to the right
D. The calcium ion (Ca^{2+}) from CaF_2 reacts with NaF, forming a precipitate

28. In the titration of a weak acid (HA) with a strong base (OH^-), at what point is the concentration of A equal to the concentration of HA?

A. At the equivalence point
B. At half the equivalence point
C. Before the equivalence point
D. after the equivalence point

29. A 25.0 mL solution of H_2SO_4 requires 50.0 mL of 0.2 M NaOH for neutralization. What is the molarity of the sulfuric acid solution?

A. 0.1 M
B. 0.2 M
C. 0.4 M
D. 0.8 M

30. Which of the following salts will undergo hydrolysis to form a basic solution?

A. NaCl
B. K_2SO_4
C. Na_3PO_4
D. NH_4Cl

Chemistry Answer Sheet G. 12 **Time allotted: 1:20 hr**

Name Hanna Ayalew Sec. 12th C No. 24

1. C	11. B	21. B
2. C	12. A	22. A
3. A	13. A	23. B
4. A	14. B	24. B
5. D	15. B	25. B
6. C	16. B	26. C
7. A	17. B	27. A
8. B	18. C	28. B
9. C	19. A	29. B
10. B	20. B	30. D C

29
30

Excellent